3.5 Completing the Square

Name _____ Period ____ Date _____

Algebra II 3.5 · Student sheet

I can…

- solve equations using square roots when one side is a perfect square.
- complete the square and solve quadratic equations.
- complete the square to write quadratic functions in vertex form.

Vocabulary · fill in the missing word

Completing the square — _____ $(b/2)^2$ to $x^2 + bx$ to make a perfect square trinomial: $x^2 + bx + (b/2)^2 = (x + b/2)^2$

Part 1 · Build the square

Sketch an x^2 square and the x-strips, split in half on two sides. How many unit squares fill the corner?

$x^2 + 6x$	$x^2 + 8x$	$x^2 - 4x$

Expression	Add	Square
$x^2 + 6x$		
$x^2 + 8x$		
$x^2 - 4x$		

Part 2 · Notes: fill in the blanks

To complete the square for $x^2 + bx$, add _____.

When solving, add it to _____ sides.

If $a \neq 1$, first _____.

Part 3 · Examples (work them with the class)

Example 1

Find the value of c that makes $x^2 - 14x + c$ a perfect square trinomial. Write it as a square.

Example 2

Solve $x^2 - 4x - 3 = 0$ by completing the square.

Example 3

Solve $2x^2 + 8x + 10 = 0$ by completing the square.

Example 4

Write $y = x^2 - 10x + 22$ in vertex form. Give the vertex.

Example 5

The height h (in feet) of a water balloon t seconds after it is launched is $h = -16t^2 + 16t + 3$. Find the maximum height.

Part 4 · Partner practice

Partner 1

Solve $x^2 + 8x + 1 = 0$ by completing the square.

Partner 2

Write $y = x^2 + 6x + 5$ in vertex form and give the vertex.